RAG Semantic Test Suite (150 Moderate-Difficulty Cases)

Purpose

This document contains 150 test inputs for the RAG security audit model. The phrasing in these test cases differs from the foundational knowledge base to test the model's semantic understanding and contextual matching capabilities.

Part 1: Expected Classification - Negative (1 - 50)

Feed these inputs into the RAG model. It should classify them as Negative/Red Flags based on the knowledge base.

1. "Only addresses mapped in the `authorizedSellers` variable are permitted to execute the `transferFrom` function via the DEX router."
2. "The administrators hold the right to append any user's wallet to the `restrictedEntities` mapping, which prevents token movement."
3. "The `updateFees` function allows the deployer to set the `marketingFee` up to a maximum value of 10,000 basis points (100%)."
4. "The `maxTxAmount` variable can be modified by the admin down to 0 wei, effectively pausing the network."
5. "The smart contract enforces a delay between trades, and the `setDelay` function allows the owner to set this to `type(uint256).max` ."
6. "The deployer can reduce the `maxWalletHolding` limit at any time, rendering users unable to receive airdrops or buy tokens."
7. "The standard ERC20 `approve` interface has been modified to include `require(msg.sender == deployerAddress)` ."
8. "During the `_transfer` execution, the `amount` is deducted from the sender, but the recipient receives `amount - hiddenFee` , with the remainder unaccounted for in state variables."
9. "The contract owner is set to `address(0xdead)` , but an undocumented `require(tx.origin == 0x123...)` exists in the administrative modifier."
10. "A public function `generateTokens(address to, uint256 amount)` is protected only by the `onlyOwner` modifier and lacks a hard cap."
11. "In the event of network instability, the multi-sig may trigger `withdrawAllStakedAssets()` to move user principal to the developer treasury."
12. "The proxy implementation can be pointed to a new logic contract via `upgradeProxy()` without triggering a timelock delay."
13. "The `releaseLP()` function can be called by the admin before the `block.timestamp` reaches the designated `unlockTime` ."

14. "A fallback function `receive()` redirects all incoming native tokens to an unverified external address rather than the protocol treasury."
15. "The contract contains a `destroyTokens(address target, uint256 amount)` function callable only by the administrative role."
16. "The `executeTransaction` function takes a `bytes` payload and uses `target.delegatecall(payload)` under an `onlyAdmin` modifier."
17. "The fallback mechanism includes a `selfdestruct(payable(admin))` command to gracefully deprecate the contract."
18. "While the total supply is defined as 1 million, the `expandSupply()` function allows the owner to bypass this theoretical limit."
19. "The contract sends ETH to the user via `msg.sender.call{value: amount}("")` before setting `userBalances[msg.sender] = 0` ."
20. "The `withdrawStakedETH` function does not utilize the OpenZeppelin `nonReentrant` modifier."
21. "The yield farming contract and the lending contract share user states, but only the lending contract checks for re-entrancy, leaving the farm exposed."
22. "The internal exchange rate is determined by calling `getReserves()` directly from the Uniswap V2 pair during execution."
23. "The protocol utilizes `token.transfer(user, amount);` but does not wrap it in a `require` statement to check for a boolean success."
24. "The `permit` function processes `v, r, s` signatures but fails to include the `block.chainid` in the typed data hash."
25. "Authentication for the `claimRewards` function relies entirely on checking if `tx.origin` matches the registered user."
26. "The internal arbitrage function hardcodes `amountOutMin` to 0 when routing swaps through external liquidity pools."
27. "Withdrawal penalties are algorithmically tied to the protocol's health factor and may scale up to 85% during periods of high stress."
28. "A daily operational fee is deducted directly from the staking vault, and the terms do not specify a maximum percentage for this deduction."
29. "Upon unstaking, user USDC is automatically swapped for the protocol's native governance token at a rate determined by the team."
30. "The `calculateYield` library automatically deducts 5% of the gross profit and routes it to the developer fund without logging an event."
31. "The smart contract interface incorrectly parses USDC inputs as having 18 decimal places instead of 6, resulting in inflated administrative fees."
32. "The Company explicitly retains the sole discretion to terminate your access to the application and seize assets without prior warning."

33. "You agree that the developers shall not be held liable for any loss of funds, even in the event of proven internal theft or gross negligence."
34. "By interacting with this protocol, you waive your right to join any class-wide arbitration or class-action lawsuit."
35. "These legal parameters are subject to change at any time, and we are under no obligation to notify you when such changes occur."
36. "All legal disputes must be filed and arbitrated exclusively in the jurisdiction of the Seychelles."
37. "If we determine you have violated our acceptable use policy, we reserve the right to confiscate the digital assets in your connected wallet."
38. "Users must obtain written consent from the team before publicly discussing any technical vulnerabilities or financial losses experienced."
39. "Any trading strategies or automated bots you connect to our API become the exclusive intellectual property of the protocol."
40. "Disputes will be resolved by an arbitrator appointed exclusively by the Company."
41. "The protocol may indefinitely suspend withdrawal processing during periods of what the Company deems to be 'network maintenance'."
42. "The core team reserves the right to increase token emissions or alter the utility of the token without holding a DAO governance vote."
43. "Premium tier access will be automatically deducted from your deposited stablecoin balance every 30 days."
44. "We may share or sell your identifiable verification data and transaction history to third-party affiliates for marketing purposes."
45. "The protocol is not legally obligated to honor the redemption value of the native utility token."
46. "When a user's collateral is liquidated, the entirety of the liquidation penalty is routed to the developer treasury rather than the liquidity providers."
47. "The interface applies a 25% premium to estimated network gas costs and retains the unspent difference as platform revenue."
48. "A 1% administrative processing fee is applied to all cross-chain bridges, which is not mentioned in the official GitBook documentation."
49. "All tokens purchased via the liquidity pool are subject to a mandatory 7-day holding period before they can be transferred."
50. "Purchasing the token is permissionless, but users must submit a KYC application to the team before they are authorized to sell."

Part 2: Expected Classification - Neutral (51 - 100)

Feed these inputs into the RAG model. It should classify them as Neutral/Standard Boilerplate based on the knowledge base.

51. "The decentralized application and accompanying smart contracts are provided on an 'as-is' basis without any warranties of merchantability."
52. "The maximum liability of the protocol for any service interruption shall not exceed the fees paid by you in the 12 months prior to the incident."
53. "These terms and conditions shall be governed by and construed in accordance with the laws of the State of Delaware."
54. "Any claim arising from the use of this software shall be settled by binding arbitration administered by the LCIA."
55. "We shall not be held responsible for delays or failures in performance resulting from acts of God, war, or severe network outages."
56. "If any provision of this agreement is deemed unlawful, that provision shall be severed without affecting the validity of the remaining provisions."
57. "This document constitutes the entire agreement between the user and the protocol regarding the use of the interface."
58. "We may update these terms from time to time, provided we give a standard written notice via our official channels."
59. "The Company may transfer or assign its rights under these terms to a parent company or acquirer in the event of a merger."
60. "These terms are specifically between you and the protocol; they do not confer any rights to any third-party beneficiaries."
61. "The web interface utilizes third-party infrastructure to verify that connecting wallets do not belong to legally sanctioned entities."
62. "We collect anonymized, aggregated behavioral data to improve the user interface and overall platform experience."
63. "Identity verification is only required if you attempt to use the fiat on-ramp features provided by our third-party payment processors."
64. "Due to regulatory requirements, access to the frontend interface is geographically blocked for residents of the United States."
65. "For users residing in the European Union, we fully comply with GDPR standards regarding your cookie preferences."
66. "Server logs containing non-identifiable web traffic data are retained for a maximum of 90 days before automatic deletion."
67. "Our frontend uses essential session cookies solely to maintain your connection state while navigating the application."
68. "You acknowledge that you are entirely responsible for calculating and remitting any taxes applicable to your cryptocurrency transactions."

69. "You must be at least 18 years of age, or the age of majority in your region, to interact with our smart contracts."
70. "Using the interface to engage in money laundering, terrorism financing, or the spread of malicious software is strictly prohibited."
71. "The core developers reserve the right to deprecate the hosted frontend website with a 30-day prior public notice."
72. "If the primary web interface is taken offline, users remain responsible for interacting with the blockchain to unwind their active positions."
73. "The contracts import a pausable module that allows the multisig to halt deposits and borrows during a recognized security incident."
74. "The core logic contracts are deployed behind an upgradeable proxy pattern to facilitate future bug fixes and feature additions."
75. "A designated admin address holds the authority to update frontend text variables and non-critical operational parameters."
76. "The protocol fetches live pricing data for all supported synthetic assets exclusively from the Chainlink decentralized oracle network."
77. "The interface will automatically estimate the required network gas and add a small buffer to ensure your transaction does not fail."
78. "Liquidity pool swap fees are algorithmic and will dynamically adjust between 0.1% and 1.5% based on recent trading volume."
79. "The frontend may be periodically unavailable due to scheduled server maintenance or necessary infrastructure upgrades."
80. "To ensure server stability, requests to our public API are limited to 100 calls per minute per IP address."
81. "Nothing published on our documentation or interface should be construed as professional financial advice."
82. "Interacting with experimental DeFi protocols carries significant risk, including the total loss of your deposited principal."
83. "You understand and agree that the value of digital assets is highly volatile and unpredictable."
84. "The protocol provides past performance metrics for reference but makes no guarantee regarding future yields or profitability."
85. "This version of the application is currently in beta; users should expect minor visual bugs or intermittent display issues."
86. "We do not endorse and are not liable for the privacy practices of any third-party websites linked within our application."
87. "You are solely responsible for the safekeeping of your private keys, seed phrase, and hardware wallet devices."

88. "We do not have custody or administrative control over the unhosted wallets you use to interact with our platform."
89. "During times of high activity, the underlying blockchain may experience congestion, leading to delayed transaction confirmations."
90. "The native governance token is intended solely for utility within the protocol and does not represent an equity stake."
91. "The smart contracts implement standard ERC-20 and ERC-721 interfaces for fungible and non-fungible asset handling."
92. "Before interacting with the vault, you must sign an approval transaction granting the contract permission to move your tokens."
93. "Token holders have the ability to submit proposals to the governance forum for community discussion and off-chain signaling."
94. "To accrue protocol emissions, users are required to deposit and lock their governance tokens into the staking contract."
95. "By providing liquidity to the decentralized exchange, you expose yourself to the mathematical risk of impermanent loss."
96. "You acknowledge that holding assets in a liquidity pool may yield a lower fiat value than holding the individual tokens in your wallet."
97. "Cross-chain transfers rely on external bridge finality, which may take up to 7 days depending on the source network's security model."
98. "Users seeking to run a validator node must meet strict hardware specifications and maintain a 99.9% uptime metric."
99. "If your validator double-signs a block or experiences prolonged downtime, your staked tokens will be subject to a slashing penalty."
100. "Our standard service level agreement dictates that non-critical support tickets will receive a response within 3 to 5 business days."

Part 3: Expected Classification - Positive (101 - 150)

Feed these inputs into the RAG model. It should classify them as Positive/Best Practices based on the knowledge base.

101. "Every public function that modifies the contract's state is wrapped with the `nonReentrant` modifier from OpenZeppelin."
102. "The smart contract includes rigorous `require(target != address(0))` checks to prevent accidental transfers to the burn address."
103. "The codebase utilizes the `SafeERC20` library, ensuring that tokens failing to return a boolean value do not cause silent failures."
104. "Any upgrade to the proxy contract is enforced by a mathematically hardcoded 48-hour timelock delay."

105. "The architecture strictly adheres to the Checks-Effects-Interactions pattern, updating all balances before executing external calls."
106. "To calculate collateral ratios safely, the protocol utilizes a Time-Weighted Average Price (TWAP) oracle rather than spot pricing."
107. "Administrative permissions are modularized using `AccessControl`, ensuring no single wallet possesses omnipotent owner rights."
108. "All off-chain signature verifications incorporate standard domain separators and type hashes compliant with EIP-712."
109. "The deployer contract includes a publicly callable `renounceRole` function allowing the initial founders to step down permanently."
110. "The protocol's GitHub repository includes a comprehensive unit testing suite that covers 98% of the smart contract logic."
111. "The terms expressly guarantee that users maintain complete self-custody of their assets and the protocol has no technical ability to freeze them."
112. "Users will be notified via our primary communication channels at least 60 days before any material changes to these terms are enforced."
113. "An immutable, fully decentralized version of the interface is permanently hosted on IPFS for uninterrupted user access."
114. "We adhere to a strict opt-in policy for data collection; absolutely no cookies or tracking scripts are loaded without explicit user consent."
115. "Users possess the right to contact us to invoke their right to be forgotten, resulting in the permanent deletion of their off-chain profile data."
116. "Our privacy documentation is intentionally drafted in plain English to ensure complete transparency without confusing legal jargon."
117. "The protocol categorically guarantees that it does not embed third-party marketing pixels or tracking beacons anywhere on the platform."
118. "A self-service portal allows users to instantly download a comprehensive CSV of their entire account history without contacting support."
119. "Alongside the binding legal text, we provide simple, non-binding summaries to help everyday users understand their rights."
120. "The `setMaxFee` function includes a `require(fee <= 200)` statement, permanently capping any potential protocol fee at 2%."
121. "Changes to emission schedules or treasury allocations can only be executed if a DAO governance vote reaches a 51% quorum."
122. "The smart contracts explicitly guarantee that zero exit fees or withdrawal penalties will ever be applied to user principal."
123. "All calculations regarding protocol revenue are completely transparent and verifiable directly on the blockchain."

124. "Token distribution was executed via a verifiable fair launch, with zero pre-mined allocations assigned to the development team."
125. "All deflationary mechanics and token burns are handled purely by immutable smart contract logic, not manual team interventions."
126. "The addresses for all developer, marketing, and protocol treasury wallets are publicly documented and tracked on Etherscan."
127. "The platform integrates with a decentralized oracle to provide real-time, cryptographic proof of reserves for all backed assets."
128. "The initial protocol liquidity pool has been cryptographically locked via Unicrypt for a guaranteed period of 24 months."
129. "Tokens allocated to early investors and core team members are locked in a public vesting contract with a 12-month cliff."
130. "To prevent denial-of-service vulnerabilities, the contract requires users to manually pull their rewards rather than pushing funds out in a loop."
131. "The protocol maintains a heavily funded bug bounty program on Immunefi to incentivize responsible disclosure of vulnerabilities."
132. "Execution of critical protocol upgrades requires authorization from a decentralized 5-of-7 Gnosis Safe multi-signature wallet."
133. "A portion of the protocol treasury is dedicated to funding an active insurance policy on Nexus Mutual to protect user deposits against exploits."
134. "The contract features a rate-limiting circuit breaker that caps the total percentage of TVL that can be withdrawn in a single 24-hour period."
135. "In the event of a dispute, users have the option to escalate the matter to Kleros, a decentralized blockchain-based arbitration court."
136. "The identities of all core contributors have been thoroughly vetted and cryptographically verified by a reputable KYC auditing firm."
137. "All smart contracts, frontend applications, and backend indexers are fully open-source and available for public review."
138. "Prior to deploying on mainnet, the codebase was subjected to three independent security audits by industry-leading firms."
139. "The core mathematical logic responsible for calculating liquidations has undergone rigorous formal verification."
140. "Governance decisions are executed via transparent, on-chain voting mechanics rather than opaque off-chain snapshot polls."
141. "Users can seamlessly delegate their voting weight to active community delegates without transferring custody of their underlying tokens."
142. "The governance implementation programmatically reverts any passed proposals if the minimum voter participation quorum was not met."

143. "The development team does not possess any admin keys or veto privileges capable of overturning a successful community vote."
144. "The entirety of the protocol's generated fee revenue is routed directly to the DAO treasury, which is managed exclusively by token holders."
145. "We maintain completely open and uncensored community forums to encourage transparent debate and constructive criticism."
146. "The DAO treasury funds a transparent grant program specifically designed to onboard independent developers into the ecosystem."
147. "The documentation provides clear, permissionless instructions enabling any individual to operate a network node."
148. "The automated market maker allows for the permissionless creation of liquidity pools for any newly launched asset."
149. "The terms of service explicitly state that the core protocol layer will never censor, alter, or revert cryptographically valid transactions."